

SUMMARY

AI Engineer with hands-on experience developing and operationalizing end-to-end AI solutions across classical ML, deep learning, and generative AI, covering everything from data preparation to production deployment of RAG pipelines, LLM-powered applications, and model inference systems.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
GenAI & LLMs: RAG, Vector DBs, Prompt Engineering
Databases: MySQL, PostgreSQL, SQLite, MongoDB
DevOps & Tools: Docker, FastAPI, REST APIs, Git, CI/CD

EXPERIENCE

AI Research Assistant

Eötvös Loránd University (ELTE)

Sep. 2025 – Present

Budapest, Hungary

- Build and optimize DP-SGD federated learning pipelines for privacy-preserving synthetic data generation ($\epsilon = 1$), engineering data preparation and feature workflows to benchmark privacy-utility trade-offs
- Author and teach the open-source IDaSec Practicum (8 labs), documenting AI architectures and best practices in responsible AI
- Collaborate with cross-functional teams and mentor graduate students in AI engineering and MLOps workflows

Data Analyst

Djezzy

Dec. 2024 – Mar. 2025

Khenchela, Algeria

- Analyzed subscriber telecom datasets using Python and SQL to identify usage trends and customer behavior patterns
- Built automated data preparation and feature engineering pipelines, facilitating data-driven decision making and reporting
- Collaborated with analytics and cross-functional teams to model subscriber metrics and optimize customer retention strategies

PROJECTS

CASPER

Python, Differential Privacy, RAG, T5, MedCPT, Docker

2026

- Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
- Achieved technical Recall@10 of 0.91 at $\epsilon \approx 44.28$, $\delta = 10^{-5}$, with 50% MIA AUC (random-guessing), demonstrating formal and empirical privacy guarantees

MyVellum

FastAPI, Groq API, LLM Orchestration, RAG, Docker, CI/CD

Dec. 2025 – Present

- Architected an LLM-powered app with modular prompt engineering, vector-backed RAG pipelines, and retrieval optimization
- Built automated evaluation gates into CI/CD for relevance and quality metrics; integrated Groq API and FastAPI with Docker

Context-Aware Access Control for IoT Device

FastAPI, Python, Multi-Agent LLMs, Docker, Wazuh, Prometheus

2026

ITAT'2026, WAS

- Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter (< 1 ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
- Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts

EDUCATION

PhD in Informatics (GPA: 5/5)

Eötvös Loránd University (ELTE)

Sep. 2025 – Expected 2029

Budapest, Hungary

Thesis: Applied synthetic data generation for secure, privacy-preserving AI systems (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence – Ranked 1st in Class (GPA: 3.7/4.0)

University of Khenchela

Sep. 2023 – Jun. 2025

Khenchela, Algeria

Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

B.Sc. in Mathematics and Computer Science – Ranked 1st in Class (GPA: 3.9/4.0)

University of Khenchela

Sep. 2020 – Jun. 2023

Khenchela, Algeria

Final Year Project: A learning management system for Algerian students.

AWARDS & CERTIFICATIONS

- Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE Sep. 2025
- 2nd Place — Starthon'25 Hackathon:** University of El Oued, Algeria Apr. 2025
- Software Engineering (Back-end Specialization):** ALX Africa (savanna.alxafrica.com/certificates/yNTLX6BYHR) Feb. 2025
- AI Career Essentials (AiCE):** ALX Africa (savanna.alxafrica.com/certificates/YEM9LC5BNm) Jul. 2024